

PowerShell Empire Post Exploitation

Professional Case Report 16

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Report date	24. septembar 2026
Case ID	CSC-2026-16
Discipline	Adversary simulation
Overall risk	Critical
Evidence status	Laboratory methodology documented

Purpose

Understand post-exploitation workflow and identify defensive telemetry that exposes PowerShell-based agent activity.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

PowerShell Empire was used in an isolated lab to demonstrate host enumeration, privilege-escalation checks and credential-access modules. The lab connected offensive workflow to defensive evidence. PowerShell monitoring is most effective when combined with privilege hygiene and credential controls.

Case Profile

Scenario	PowerShell Empire was used in an isolated lab to demonstrate host enumeration, privilege-escalation checks and credential-access modules.
Environment	Kali Linux Empire server and Windows training target.
Tools	PowerShell Empire, PowerUp, PowerShell, Windows event logging, Credential modules

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Started Empire server and client components.
2. Generated a controlled stager and delivered it to the target.
3. Executed host-enumeration modules to understand context.
4. Ran privilege-escalation checks and reviewed findings.
5. Examined credential module behavior without retaining sensitive data.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	Living-off-the-land execution	High	PowerShell can host complex post-exploitation logic while appearing similar to administration activity.
F02	Privilege-escalation discovery	Critical	Automated checks identify misconfigurations that can raise an agent from user to administrator.
F03	Credential access	Critical	Credential modules can expose hashes or secrets that enable lateral movement.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

```
powershell-empire server
powershell-empire client
usemodule powershell/situational_awareness/host/winenum
usemodule powershell/privesc/powerup/allchecks
usemodule powershell/credentials/powerdump
```

Risk Assessment

Risk	Level	Operational impact
Credential compromise	Critical	Secrets can be reused across systems
Privilege escalation	Critical	Local weaknesses can provide administrator control

Recommendations

1. Enable Script Block Logging, Module Logging and AMSI visibility.
2. Remove local privilege-escalation misconfigurations.
3. Use credential protections and unique local administrator passwords.
4. Alert on Empire-related command patterns and unusual PowerShell parents.

Knowledge Demonstrated

- Post-exploitation workflow
- PowerShell telemetry
- Privilege assessment
- Credential-risk analysis

Conclusion

The lab connected offensive workflow to defensive evidence. PowerShell monitoring is most effective when combined with privilege hygiene and credential controls.

Hands On Lab Practical 16

Completed as part of the authorized cyber security course practical work.